



Fatty acid esterification by reactive distillation. Part 1: equilibrium-based design

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Abstract

Esters of fatty acids are currently produced in batch processes. In this study we present an innovative process based on reactive distillation (RD). The synthesis of a several fatty esters is possible in the same RD set-up, making possible the development of a continuous multipurpose process. Conceptual design is presented as a systematic methodology based on thermodynamic analysis combined with computer simulation. This approach sets also targets for limited experimental work. The methodology is illustrated by the esterification of lauric acid with 2-ethylhexanol and methanol, the heaviest and lightest alcohol in the C_1 – C_8 series. The first part presents the design of a RD set-up based on chemical equilibrium. Both reactions can be accommodated in the same hardware, but with different operation procedures. The alternative with alcohol reflux is suitable for heavy alcohols that form heterogeneous azeotropes with water. Another alternative with acid reflux can accommodate both light and heavy alcohols, and may be seen as a generic process for fatty acid esterification.

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1. Introduction

Fatty esters are important fine chemicals used in the manufacturing of cosmetics, detergents and surfactants. They are currently produced in batch reactors by homogeneous catalysis. Batch processes are penalised by important shortcomings. Firstly, a costly post-treatment by distillation is necessary for the reuse of excess alcohol and product purification. Secondly, the catalysts, often sulphuric or *p*-toluenesulphonic acid is lost, give rise to secondary reactions and generate pollution problems. Organic acidic ion-exchange resins could be applied, but these are not suitable at temperatures higher than 413 K. Alternatively, solid acid catalysts, as those based on sulphated or tungstated zirconia may provide sufficient thermostability.

Catalytic reactive distillation (RD) for fatty esters synthesis seems to be attractive for both economical and ecological reasons. A major advantage is that the counter-current flow allows an excess of reactant alternatively at the top and

the bottom of the reactive zone, resulting in a more uniform and higher reaction rate. Equilibrium reactions are excellent candidates for RD, as in case of well-known processes for methyl acetate or MTBE. It is difficult to reconcile reaction and separation conditions, satisfying simultaneously strong constraints regarding thermodynamics, chemistry and kinetics. Hence, the development of real processes is by far more complicated than is suggested by the application of some generic algorithmic tools. A major obstacle is finding a suitable catalyst, as was also noticed by Tuchlenski et al. (2001).

The goal of this study is to investigate systematically the possibility of applying RD to the synthesis of a range of fatty acid esters. As representative species we consider lauric acid, and two alcohols, 2-ethyl-1-hexanol (iso-octanol), and methanol, the highest and the lowest boiler in the series C_1 – C_8 . Note that these are today the most manufactured fatty esters in industry. Particularly, methyl ester could play a significant role in the future as a major component of ‘bio-diesel’ fuels.

This research is presented on the space of two articles. The first part deals with the general methodological frame, a comprehensive thermodynamic study, and the development

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of design alternatives based on the assumption of chemical equilibrium. The second article discusses the kinetics-based design using sulphated zirconia catalyst. On this basis the column configuration is optimised.

2. Previous work

A continuous RD based process applied to fatty acids has been described recently by Bock, Wozny, and Gutsche (1997) for the manufacture of isopropyl myristate. The set-up consists of two columns. The process makes use of an excess of alcohol, which distillates together with water as top product. The alcohol is recycled as azeotrope with water after separation in a second column. The acid catalyst is lost, neutralised and washed from the final product.

RD is considered a major direction in process intensification. Efforts have been made to arrive at systematic design methods, mainly by incorporating the chemical reaction in the prolific residue curve map (RCM) technology. Some observations can be made. Firstly, the analysis of multi-component mixtures with more than three components by graphical representation is tedious. Transformed variables, as proposed by Barbosa and Doherty (1987) offer the possibility of 2-D representation of a quaternary mixture at chemical equilibrium. Secondly, the columns' sequencing from RCM at infinite reflux might be far from the finite reflux design, particularly for strong non-ideal systems. Thirdly, the combination of chemical reaction with phase separation requires reliable data, not always available in the databases of simulation software. That is why mastering the simultaneous chemical and phase equilibrium (CPE) is a pre-requisite of any conceptual development.

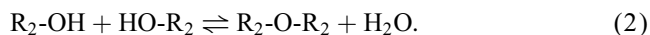
Different methods and algorithms for solving CPE were described in literature, as by Castier, Rasmussen, and Fredenslund (1989), Xiao, Zhu, and Yuan (1989), Ung and Doherty (1995a, b), Cisneros, Gani, and Michelsen (1997), Wasykiewicz and Ung (2000). Aspen PlusTM (2001) offers a reliable algorithm for the computation of the equilibrium composition by Gibbs free energy minimisation taking into account three-phase equilibrium.

3. Definition of the process synthesis problem

Fatty acid esters are produced by the reversible reaction:



An undesired secondary reaction that could take place is the etherification of alcohols:



Further impurities may result from alkene formation and dimerisation at higher temperatures. As typical substrates we consider lauric (dodecanoic) acid, as well as 2-ethylhexanol and methanol, the heaviest and the lightest component in the C₁–C₈ series.

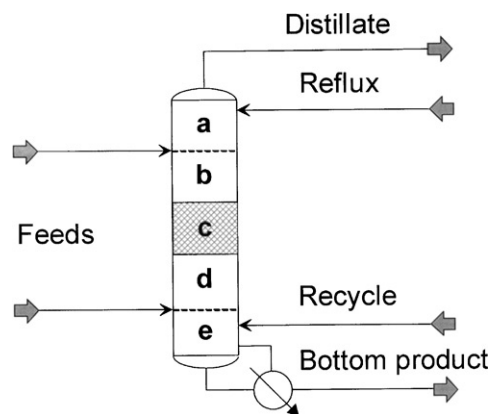


Fig. 1. Conceptual set-up of a RD column operated in counter-current of reactants.

Because water is by-product, the segregation into aqueous and organic phase is very likely to occur. On the other hand water has to be removed continuously and selectively from the system, in order to shift the chemical equilibrium to total conversion. Therefore, the advanced removal of water from the liquid phase is required in designing an RD process. In addition, the thermodynamic behaviour of the reaction system is very non-ideal, depending strongly on the couple alcohol–water.

The design objective is a process for high purity fatty esters, over 99.8%, suitable for applications in cosmetics. High purity initial reactants are assumed, so that troublesome impurities could come only from secondary reactions.

An essential feature is that the RD process fatty esters will make use of a solid catalyst with acidic properties. However, the use of a solid catalyst imposes constraints regarding the reaction conditions in which the catalyst keeps its activity and selectivity over a longer period.

Another distinctive feature of a continuous RD design is that the reactants should be fed in stoichiometric ratio. Unlike in batch processes, an excess of reactant should be realised mainly by internal recycle, and only in a very limited manner by external separation. If auxiliary equipment is required, this should be simple and low cost.

The conceptual layout of an RD set-up for fatty esters synthesis is presented in Fig. 1. Consider the reaction as $R_1 + R_2 \rightleftharpoons P_1 + P_2$, in which R_1 = alcohol, R_2 = fatty acid, P_1 = water and P_2 = fatty ester. The mixture is highly non-ideal in nature and the components form different azeotropes.

In general, five zones may be distinguished, starting from the top, as follows:

- Rectifying section: separation of the light product P_1 from the heavy reactant R_2
- Recovery of un-reacted light reactant R_1 by absorption in heavy reactant R_2
- Reaction zone between R_1 and R_2 in counter-current flow

- (d) Recovery of untransformed heavy reactant R_2 by stripping with light reactant R_1
- (e) Stripping section: separation of the heavy product P_2 from the light reactant R_1 .

Acid is fed at the top of the reaction zone, preferably heated up to the reaction temperature. Alcohol enters as vapour at the lower location of the reaction zone. The RD column may contain stripping and rectification segments. Fatty ester is in this case always the heaviest component, and by consequence it is obtained as bottom product. The RD device should also ensure a quantitative separation of water in the top.

Following Fig. 1 the only constraints for feasibility are: $\alpha_{R_1/R_2} > 1$, $\alpha_{P_1/R_2} > 1$ and $\alpha_{R_1/P_2} > 1$. These constraints are satisfied by alcohols in the range C_1 – C_8 (nbp under 473 K) and fatty acids of minimum six carbon atoms (nbp above 473 K). The second constraint is fulfilled despite the presence of the azeotrope water/acid, very close to pure water on molar basis (Table 2). On mass basis the amount of acid is larger due to high molecular weight. However, the quantitative separation of water is ultimately possible since the solubility of acid in water is extremely small (Section 6.4).

Summing up, the RD device should receive reactants in stoichiometric ratio, and deliver ester and water as bottom and top products, respectively. The problem is if such device could be designed by taking into account the constraints mentioned so far.

4. Methodology

The methodology applied consists of an evolutionary search for the feasible design space by means of a systematic combination of thermodynamic analysis, computer simulation and limited experiments. Conceptual design is constructed as a logical procedure imbedding four types of activities: analysis, synthesis, evaluation and experimentation (Cisneros, Schenk, & Gani, 1996). The environment of constraints ensures a proper bounding of the feasible design space. The methodology is illustrated in the Fig. 2 and explained in more detail in Appendix A. This methodology is generic at least for similar systems. The first element of similarity is the equilibrium reaction with water as product, giving rise to the problem of L–L segregation. The second element is the similarity of thermodynamics for a class of substrates. Both fatty acids and fatty esters manifest a certain analogy in physical properties, although the alcohol–water couple may exhibit significant differences. The third element is the use of acid solid catalyst. This enables the manipulation of the reaction rate to match the separation requirements. However, the use of a catalyst adds supplementary constraints with respect to the phase-behaviour and the hydraulics of the internals.

This methodology will be illustrated by the development of a conceptual design for the esterification of dodecanoic acid with two alcohols, 2-ethylhexanol and methanol, respectively.

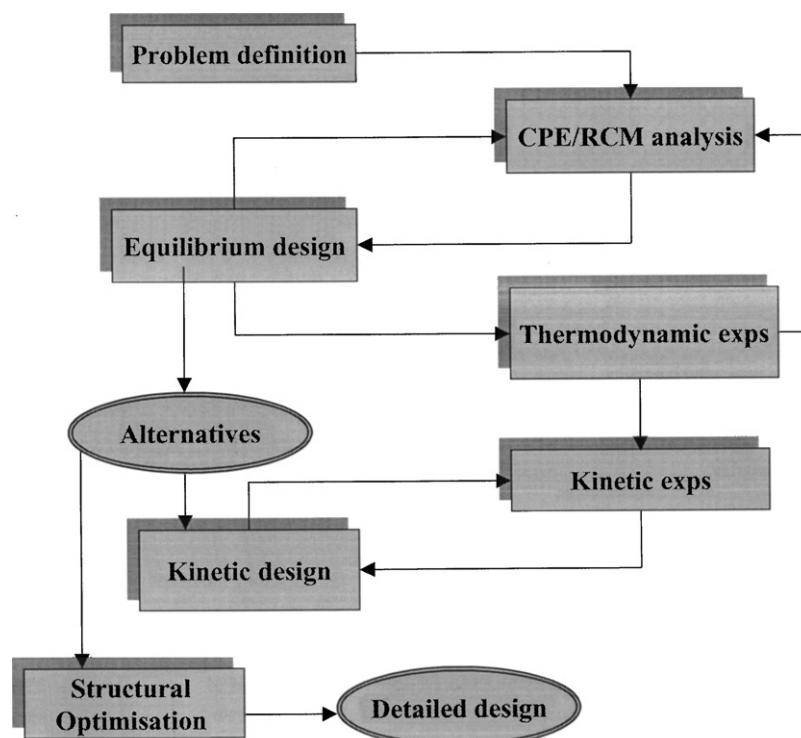


Fig. 2. Methodology for the development of an RD process.

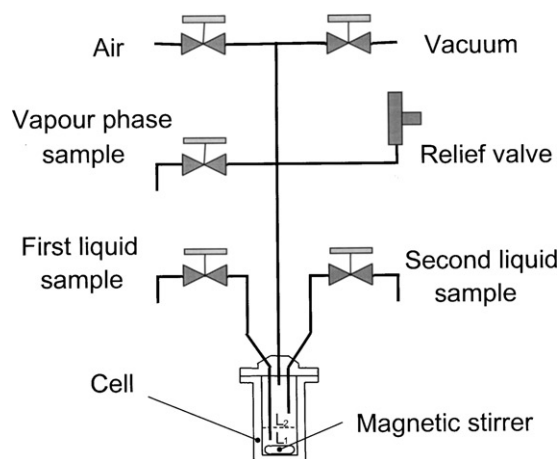


Fig. 3. Cell for CPE experiments.

5. Experimental set-up

A cell was designed for measuring simultaneous CPE over a large pressure and temperature range, up to 1 MPa and 473 K (Fig. 3). The cell has a volume of about 100 ml and is provided with glass windows for the visualisation of L–L phase segregation. The temperature is kept constant by magnetic stirring and full cell immersion in a thermostatic bath of silicone oil. Temperature was measured by a Pt-100 thermo-resistance with accuracy of ± 0.01 K. Pressure was measured by a miniaturised sensor (ENTRAN) with accuracy of ± 20 Pa. Samples of both vapour phase and two liquid phases were automatically analysed by GC: capillary column, DB-1 with internal diameter 0.2 mm and 30 m length, carrier gas: $1.9 \text{ cm}^3 \text{ min}^{-1}$ He.

6. Esterification of dodecanoic acid with 2-ethylhexanol

6.1. Data generation

Table 1 presents the boiling points of key components at normal pressure. Vapour pressure data of 2-ethylhexyl dodecanoate are not available in the literature. Estimation of normal boiling point (nbp) by group contribution methods leads to controversial values: 714 K by Joback's method (Joback & Reid, 1987) and 605 K by Gani's method (Constantinou & Gani, 1994). Note that water forms heterogeneous azeotropes with a 2-ethylhexanol and dodecanoic acid (Table 2). Remarkably, the solubility of water in any of the acid or alcohol is substantial, while the reverse

solubility, of the organic components in water, is extremely low, particularly for the acid. Pressure has little effect on the azeotropic point. No information about the water-ester azeotrope was available, but it could be anticipated that they form an immiscible system.

6.2. Preliminary chemical and phase equilibria

Inside the reactive zone, CPE occur simultaneously. The composition of vapour and liquid phases at equilibrium can be found by Gibbs free energy minimisation. Alternatively, equilibrium constant based on fugacity in vapour phase can be determined, and from this the equilibrium composition (Lee & Lin, 1999).

Table 3 illustrates the computation of equilibrium constant for the reaction lauric acid with 2-ethylhexanol in three situations, corresponding to the methods used for physical properties: (1) initial estimation with Aspen Plus data, (2) Improved estimation of ester boiling point, (3) Incorporating VLE, LLE and CPE experimental data.

Gibbs free energy of formation is known for all components, except for 2-ethylhexyl dodecanoate, which has been estimated by Aspen Plus by group contribution methods. Phase equilibrium can be calculated by means of UNIQUAC model. The interaction parameters for 2-ethylhexanol/water are available in Aspen PlusTM, the others were estimated by UNIFAC. Fig. 4 shows the variation with the temperature of K_a and K_γ , the equilibrium constant based on activity and activity coefficients, respectively, by means of the information available in Aspen Plus. It may be seen that K_a takes high values, between 10 and 50, over the range of 350 to 475 K. The correction introduced by K_γ is significant. However, values of $K_x = K_a/K_\gamma$ close to 10 can be obtained at higher temperature, leading to the conclusion that molar conversion over 98% could be possible.

The diagram in Fig. 5 shows RCM in homogeneous region and tie-lines in heterogeneous region at normal pressure and simultaneous CPE. The computation is explained in Appendix B. Note the use of special coordinates, X_1 (acid + water) and X_2 (acid + ester) that allows the representation of all four components (Barbosa & Doherty, 1987). Curve a designates the boundary between homogeneous and heterogeneous region, as calculated with data originated exclusively from Aspen. Curve b will incorporate experimental results, as explained later. At the left side there is a homogeneous organic phase, where the direction of residue curves converge clearly from reactants to the main

Table 1
Normal boiling points for key components in K

Pressure	Water	2-Ethylhexanol	Dodecanoic acid	2-Ethylhexyl dodecanoate
101.32 kPa	373.15	457.75	571.78	714 ^a , 605 ^b

^aEstimated with Aspen Plus by Joback method.

^bEstimated with Aspen Plus by Gani method.

Table 2
Azeotropes of water with 2-ethylhexanol and dodecanoic acid at 101.32 Pa

	Water (1)/2-ethylhexanol (2)	Water (1)/dodecanoic acid (2)
T_{az} (K)	372.25	373.147
y_{az}	0.9679	0.999900
$x_{1,aq}$	0.9996	0.999988
$x_{1,org}$	0.2411	0.2502

Table 3
Calculation of CPE of 2-ethylhexanol and dodecanoic acid by different methods

Parameters	Initial estimation	Improved estimation ^a	Incorporation of experimental data
2-ethylhexyl laurate			
$\Delta^f G_{298}^0$ ester (kJ mol ⁻¹)	-195.29	-195.68	-196.36
$\Delta^f G_{298}^0$ gas phase (kJ mol ⁻¹)	-11.88	-12.27	-12.95
K_f gas phase at 298 K	120.6	141.1	185.8
$T = 423$ K			
K_f	26.8	31.3	121.6
$\prod (P_i^{sat})^{v_i}$	0.15	8.67	4.25
K_a	180.4	3.61	28.62
K_γ^b	5.17	7.62	6.73
K_x^b	9.00	0.42	5.27
$T = 473$ K			
K_f	7.9	16.9	112.7
$\prod (P_i^{sat})^{v_i}$	0.17	5.28	3.18
K_a	46.5	3.20	35.5
K_γ^b	4.55	6.13	5.85
K_x^b	10.23	0.52	6.06

^aUse of predicted vapour pressure by Aspen Plus (Gani method).

^bActivity coefficients' correction calculated for equimolar reactant ratio.

product, the ester. On the other side it may be observed that water is located in the heterogeneous region. Note also the presence of a third homogeneous region (only water phase) in the right corner of the diagram, not visualised because of the scale. Because the solubility of organic components in water is so small, quasi-quantitative separation of water as by-product can be realised. Fig. 5 shows that a large region of heterogeneity exists, for which the temperature is just below 373 K. The mixing line of acid and alcohol falls in the heterogeneous region. As a consequence, the temperature in the reaction zone must exceed 373 K to avoid the formation of a second water-phase. We did not identify reactive azeotropes between the components of the reaction mixture (Okasinski & Doherty, 2000).

6.3. Preliminary conceptual design

The ∞/∞ analysis (infinite stages/infinite reflux) in the RCM based leads to the conclusion that an RD process could be feasible, as the trajectories converge to the ester vertex, which is the highest boiler, a stable node. The

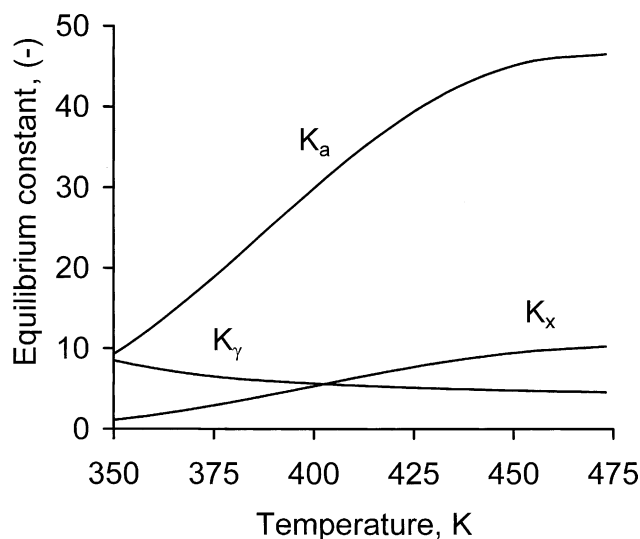


Fig. 4. Liquid phase esterification of dodecanoic acid with 2-ethylhexanol: equilibrium constants predicted by Aspen PlusTM at boiling point, starting with equimolar ratio of reactants.

hetero-azeotrope of water–alcohol forms an unstable node and the hetero-azeotrope water–acid is a saddle. In this case it is obvious that the total consumption of acid is more advantageous.

Examining the temperatures at the column ends at atmospheric pressure it is observed that these would be about 373 K in top and 713 K in bottom (Fig. 6). The latter is not acceptable, a temperature lower than 473 K being recommended to avoid thermal decomposition of the ester. VLE data for the binary ester/alcohol indicate that bottom temperature below 473 K can be realised by operating under vacuum at 32 kPa and allowing about 12 mol% alcohol in the product. This amount of alcohol can be recovered in an evaporator that serves equally for product conditioning.

The above considerations lead to the conceptual flow-sheet presented in Fig. 7. Acid and alcohol are fed counter-currently at the top and bottom of the reaction zone, respectively. Bottom product containing ester and some alcohol is sent to evaporator, from which the alcohol is recovered and recycled to the reactive zone. Large difference in boiling points allows sharp separation in one or two equilibrium stages. Top vapour containing water and some alcohol is condensed and separated by decanting. Finally, the water product can be removed quantitatively with the aqueous phase, while the alcohol-rich phase is refluxed back into the RD column.

The feasibility of above set-up was evaluated by simulation with Aspen Plus 10.2. The RD column was simulated as a stripper followed by three-phase flash, with the organic phase refluxed to the column. It was found that 3 to 5 equilibrium stages are necessary to achieve over 99% conversion when the feed temperatures was at 383 K. The stripping zone may be limited to 2–3 stages, while the rectification zone to 1–2 stages.

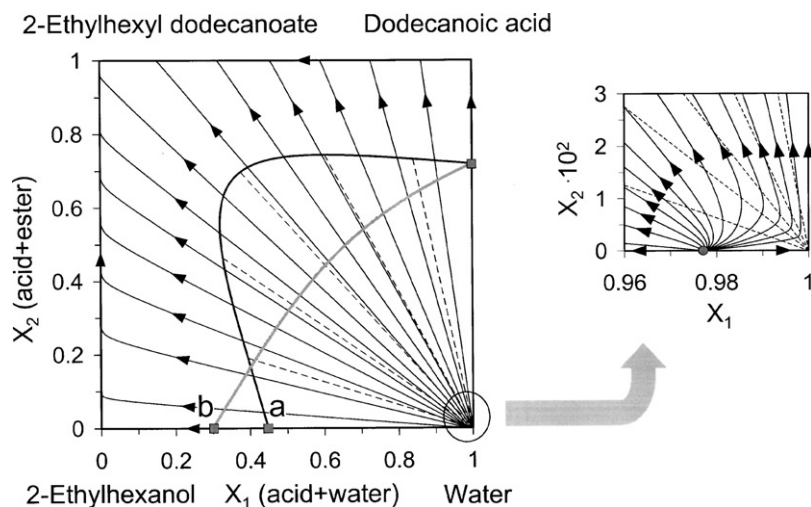


Fig. 5. Generalised CPE diagram for the esterification of dodecanoic acid with 2-ethylhexanol at 101.32 kPa showing RCM (—), LLE (---a), tie-lines (-----), solubility limits of binaries (■) and water/2-ethylhexanol azeotrope (●) for initial estimation by Aspen PlusTM. Effect of improved estimation of thermodynamic parameters on LLE (—b). The transformed coordinates are $X_1 = x_{\text{acid}} + x_{\text{water}}$, and $X_2 = x_{\text{acid}} + x_{\text{ester}}$.

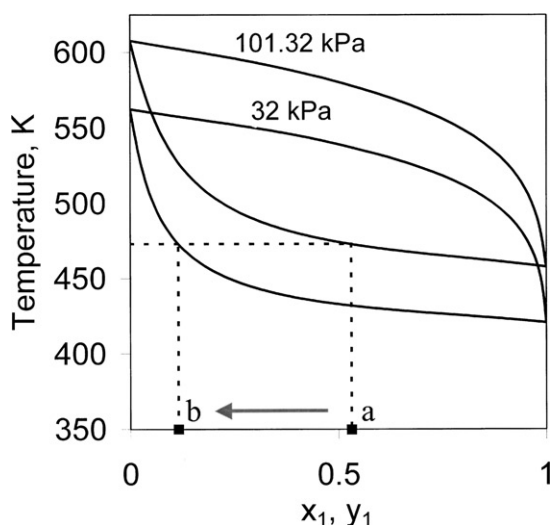


Fig. 6. T - xy diagram of the binary 2-ethylhexyl dodecanoate/2-ethylhexanol. Limiting the temperature to 473 K, the molar fraction of 2-ethylhexanol in 2-ethylhexyl dodecanoate can be reduced from about 0.5 (a) to about 0.1 (b) by decreasing the pressure.

6.4. Thermodynamic experiments

Simulation reveals that the accurate computation of simultaneous heterogeneous phase and chemical equilibrium is essential. In this case the following experimental information is necessary:

- VL(L)E for binaries involving water with alcohol and acid. If heterogeneous azeotropes are formed, the description of immiscibility region is essential. In this respect we found that using mutual solubility and azeotropic points are sufficient to describe accurately the whole range of VLLE.

- The activities of acid and alcohol. These cannot be determined directly because of chemical reaction. However, activity coefficients can be assessed by regression from CPE experiments, and correlated by liquid activity models (Lee & Lin, 1999; Lee & Liang, 1998).

Fig. 8 presents experimental vapour pressure data for 2-ethylhexyl dodecanoate. Experimental nbp is 607.6 K, a value close to the prediction by Aspen by Gani's method. Note that the estimation of the whole saturation curve by Riedel method from the normal boiling point is in large errors. Experimental vapour pressure of 2-ethylhexyl dodecanoate can be fitted over the range 450–600 K by an Antoine-type equation (see Table 4).

Solubility of 2-ethylhexyl dodecanoate in water at normal pressure on the range 300 to 360 K can be described by the following polynomial in temperature:

$$S = 1.6908 \times 10^{-5} + 1.4816 \times 10^{-7} T + 3.1158 \times 10^{-10} T^2, \quad (3)$$

where S is expressed in wt% and the temperature in K.

Given their importance, phase equilibrium data for water/dodecanoic acid and water/2-ethylhexanol have been assessed. The complete phase diagram for the binaries water/2-ethylhexanol and water/dodecanoic acid was described by UNIQUAC model. Note that the binary interaction parameters have been identified using LLE data (Stephenson, Stuart, & Tabak, 1984) plus the azeotropic point as VLE (Gmehling, Menke, Krafczyk, & Fischer, 1994). We checked that this procedure allows an accurate prediction of the liquid–liquid split, while preserving sufficient accuracy for vapour–liquid equilibrium. Presently, it is of primary interest in searching the occurrence of a water-phase, and therefore the

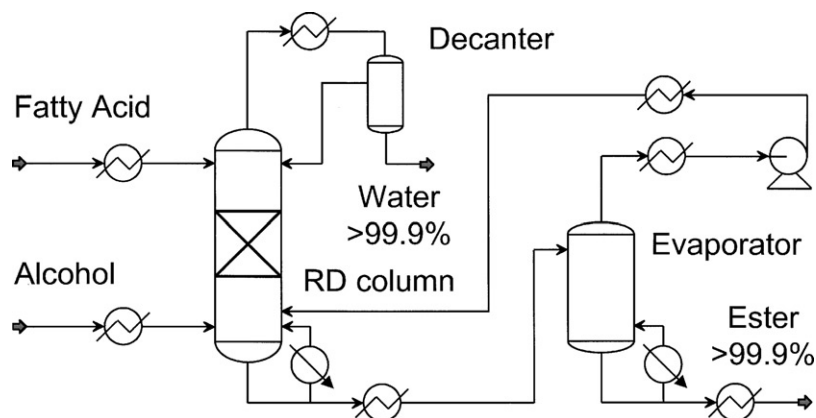


Fig. 7. Preliminary flowsheet for the synthesis of the 2-ethylhexyl dodecanoate.

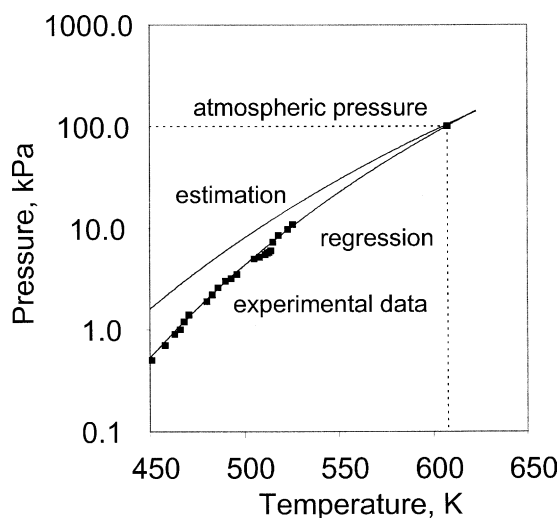


Fig. 8. Vapour pressure data of 2-ethylhexyl dodecanoate. Firstly, the vapour pressure was estimated by Riedel method using the estimation of normal boiling point (Gani). Secondly, a model based on measurements of vapour pressure from 450 to 530 K and measured boiling temperature at normal pressure was used (parameters available in Table 4).

accuracy of the thermodynamic model has to be maximised with this respect. Binary interaction parameters for UNIQUAC liquid activity model are given in the Tables 5 and Table 7.

In a second approach, the computation of CPE was reviewed by including parameters determined from experimental data. Fig. 9 presents the equilibrium constant K_x (based on molar fractions), as well as the equilibrium conversion calculated from it. Three computation methods are compared: (1) initial estimations given by Aspen, (2) 'improved' estimations using as primary information the normal boiling point of ester predicted by Gani's method, and, (3) using thermodynamic properties of 2-ethylhexyl dodecanoate and binary interaction parameters obtained by regression of experimental data (Tables 4, 5 and 7).

It can be seen that the correction introduced only by the vapour pressure of the fatty ester (method 2) is not sufficient. K_x was lower by an order of magnitude than found experimentally (Fig. 9a). The difference is also visible in the RCM plot, as curve b in Fig. 5, which indicates that the reaction trajectory could fall in a heterogeneous region. By consequence, the predicted equilibrium conversion by the second method drops to very low values of about 20–30% (Fig. 9b). Surprisingly, the equilibrium conversions calculated by the third method, in agreement with the experimental values, are not far from the initial estimation with zero experimental data! This fact merits a deeper analysis.

A consistent explanation can be found by examining the numerical values in Table 3. It may be seen that the equilibrium constant based on fugacities K_f is in the same range of values by the three methods. However, the equilibrium constant K_a based on liquid activities is very different. The difference originates from the correction for vapour pressures of the pure components. The initial estimation is particularly wrong, giving a value of K_x close to 10, and a relatively high equilibrium conversion. Contrary, the 'second' estimation gives lower values for K_x , and low equilibrium conversion. Now, the third method based on experimental data gives K_x between 5 and 6, and by consequence equilibrium conversion of about 80%. Note that the correction introduced by the constant of activity coefficients K_γ does not vary too much, so that the correction comes essentially from K_a .

A new RCM has been calculated with parameters regressed from experimental data (Fig. 10). This plot is to be compared against the CPE initial estimation given by the curve a in Fig. 5. The general aspect is preserved, but there are differences in the description of the heterogeneous region, which are significant for design. The most important is the location of the azeotropes 2-ethylhexanol/water and dodecanoic acid/water, now correctly reproduced because using interaction parameters regressed from experimental LLE.

Table 4

Parameters of the extended Antoine equation^a, over the range 450–608 K

Comp.	2-Ethylhexanol	Dodecanoic acid	Water	2-Ethylhexyl dodecanoate	Methanol	Methyl dodecanoate
A	273.62	201.56	73.649	23.578	82.718	155.18
B	18766	20484	7258.2	6247.3	6904.5	14494
C	0	0	0	−89.281	0	0
D	−36.767	−24.334	−7.3037	0	−8.8622	−18.912
E	1.98E-05	8.06-18	4.17E-06	0	7.47E-06	7.42E-06
F	2	6	2	0	2	2

^a $\ln P^{\text{sat}} = A - B/(T + C) + D \ln(T) + ET^F$ with $[P^{\text{sat}}] = \text{Pa}$ and $[T] = \text{K}$.

Table 5

UNIQUAC parameters^a for the reactive system of dodecanoic acid and 2-ethylhexanol

Comp. 1	2-Ethylhexanol	2-Ethylhexanol	2-Ethylhexanol	Dodecanoic acid	Dodecanoic acid	Water
Comp. 2	Dodecanoic acid	Water	2-Ethylhexyl dodecanoate	Water	2-Ethylhexyl dodecanoate	2-Ethylhexyl dodecanoate
A_{12}	0	11.517	0	−0.29924	0	−0.04839
A_{21}	0	−10	0	−0.38437	0	−0.44196
B_{12}	49.44	−785.57	90.824	−195.44	97.72	−203.49
B_{21}	−52.77	−2.807	−183.25	−107.62	−140.7	−738.1
C_{12}	0	−1.7187	0	0	0	0
C_{21}	0	1.5911	0	0	0	0

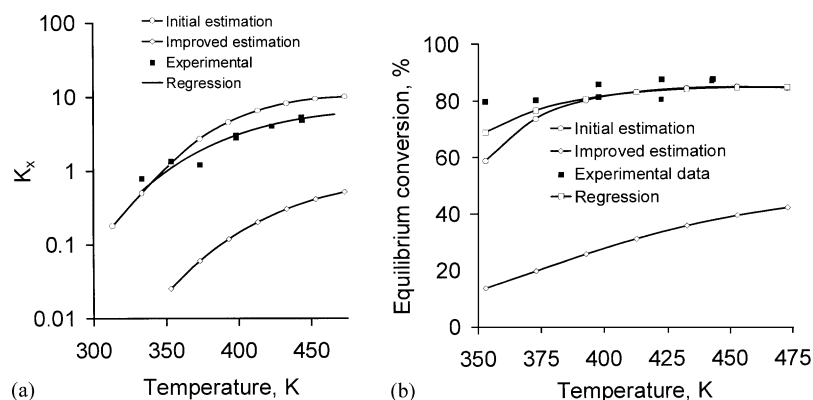
^aFor UNIQUAC expressions see Sandler (1999) with $\tau_{ij} = \exp(A_{ij} + B_{ij}/T + C_{ij}T)$ and $[T] = \text{K}$.

Fig. 9. Predicted and experimental CPE for 2-ethylhexanol and dodecanoic acid for an initial equimolar mixture: (a) equilibrium constant, and (b) equilibrium conversion. Note: *Initial estimation* refers to data available in Aspen PlusTM. *Improved estimation* refers to better estimation of nbp and VLLE. *Experimental* refers to data from composition measurements. *Regression* refers to thermodynamic parameters fitted on CPE measurements.

6.5. Reviewed conceptual design

The preliminary design is revisited taking into account the improved knowledge on thermodynamics. In principle, the design does not changed significantly. The temperature and composition profiles in the RD column (Fig. 11) indicate that the esterification of 2-ethylhexanol with dodecanoic acid requires low number of equilibrium stages and small reflux ratio. Both ester and water can be obtained at high purity in top and bottom, respectively.

7. Esterification of dodecanoic acid with methanol

7.1. Data generation

All the thermodynamic data for pure components and binary mixtures involved in the esterification of dodecanoic acid with methanol are available in Aspen Plus databank.

7.2. Preliminary chemical and phase equilibria

The thermodynamic parameters presented in Tables 4, 6 and 7 were used to calculate and draw the CPE diagram

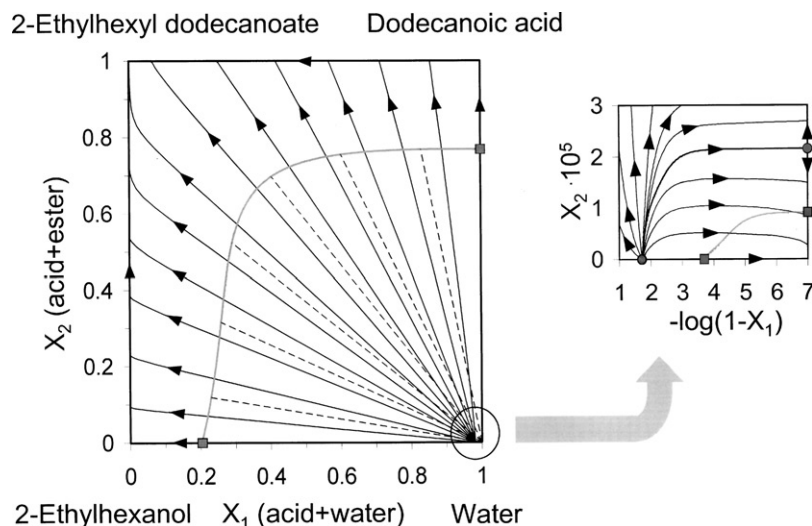


Fig. 10. Generalised CPE diagram for the esterification of dodecanoic acid with 2-ethylhexanol at 32 kPa showing RCM (—), LLE (—), tie-lines (-----), distillation line boundary (—), solubility limits of binaries (■) and azeotropes (●).

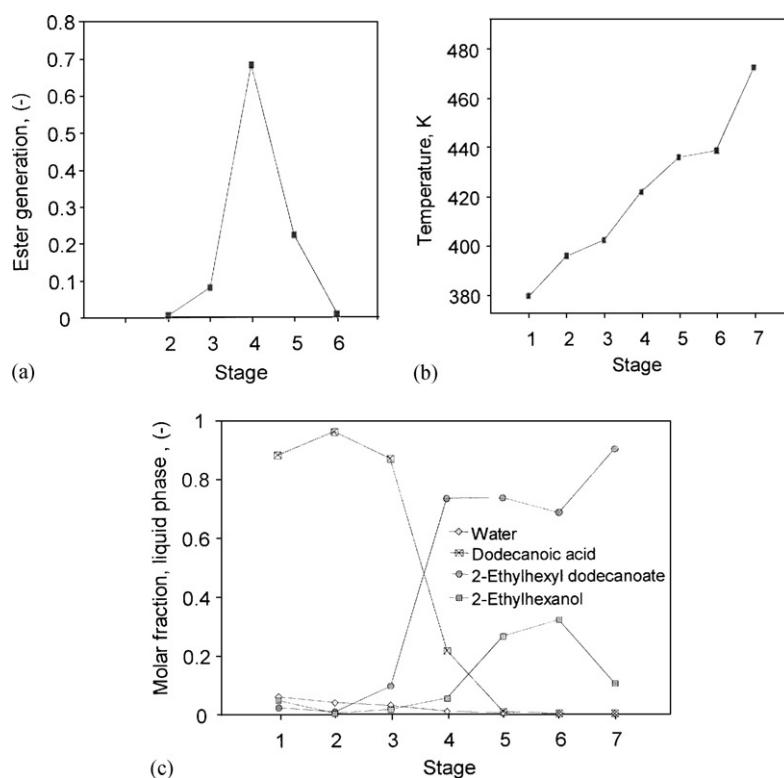


Fig. 11. Profiles in RD column for 2-ethylhexyl dodecanoate synthesis: (a) temperature profile, (b) ester generation profile, and (c) composition profiles.

for dodecanoic acid and methanol system at normal pressure. Fig. 12 shows two regions, (1) homogeneous acid/ester mixtures with excess of methanol, and (2) heterogeneous region in which a second water-phase appears. The chemical reaction, represented by a line linking the vertices of methanol and dodecanoic acid, falls in the heterogeneous region over a large reactant ratio. From separation viewpoint

methanol is an unstable node, while dodecanoic acid, methyl dodecanoate and water are stable nodes. The products are separated by two RD boundaries. In consequence, the production of high purity methyl ester from an equimolar mixture acid/alcohol is not feasible in a single column at infinite reflux and number of stages. The simulation with Aspen Plus confirms this analysis.

Table 6
UNIQUAC parameters^a for the reactive system of dodecanoic acid and methanol

Comp. 1	Methanol	Methanol	Methanol	Dodecanoic acid	Dodecanoic acid	Water
Comp. 2	Dodecanoic acid	Water	Methyl dodecanoate	Water	Methyl dodecanoate	Methyl dodecanoate
A_{12}	0	-1.0662	0	-0.29924	0	0
A_{21}	0	0.6437	0	-0.38437	0	0
B_{12}	48.3493	432.8785	31.7890	-195.44	238.8469	-216.733
B_{21}	-309.554	-322.131	-539.979	-107.62	-369.561	-658.816

^aUNIQUAC model as in Table 5 with $C_{ij} = 0$.

Table 7
Dimensionless surface (q) and volume (r) parameters^a for the UNIQUAC model

Parameters	Water	Dodecanoic acid	2-Ethylhexanol	2-Ethylhexyl dodecanoate	Methanol	Methyl dodecanoate
q	1.400	7.472	5.208	11.752	1.432	7.976
r	0.920	8.931	6.151	14.239	1.431	9.548

^aFor UNIQUAC expressions see Sandler (1999).

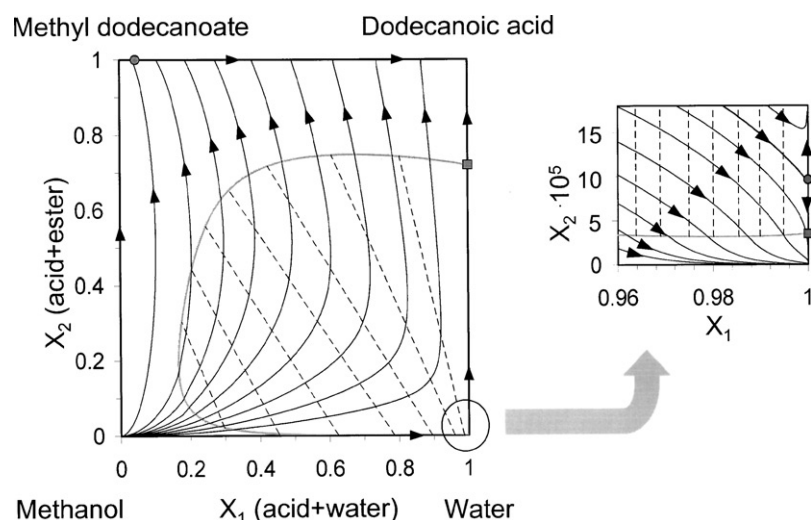


Fig. 12. Generalised CPE diagram for the esterification of dodecanoic acid with methanol at 101.32 kPa showing RCM (—), LLE (---), tie-lines (.....), azeotropes (●), and solubility limits of water/dodecanoic acid mixture (■).

On the contrary, we found by computer simulations that high purity of methyl dodecanoate might be produced in a two-column arrangement: an RD column using an excess of methanol in counter-current with dodecanoic acid, and an auxiliary distillation column for methanol recovery. Note that hardware cost and energy consumption for the alcohol recovery column might be much higher than for the main RD column.

7.3. Preliminary conceptual design

However, the ∞/∞ analysis regarding the unfeasibility of an RD process is not necessarily true at finite reflux. In

order to analyse this issue, we have realised a FORTRAN computer program for tray-by-tray computation based on the solution of MESH equations including chemical reaction with equilibrium assumption. The pressure and bottom temperature are specified at 101.32 kPa and 473 K. The computation starts from the bottom with purity specifications corresponding to molar ratio ester to acid of 999, as well as to a ratio alcohol to ester product of one. As an independent parameter we vary the boilup ratio, which gives the vapour flow rate leaving the first stage. Vapour composition is obtained from CPE, whereas liquid composition and flow rate from the upper stage come from material balance. The computation develops on a certain number of stages

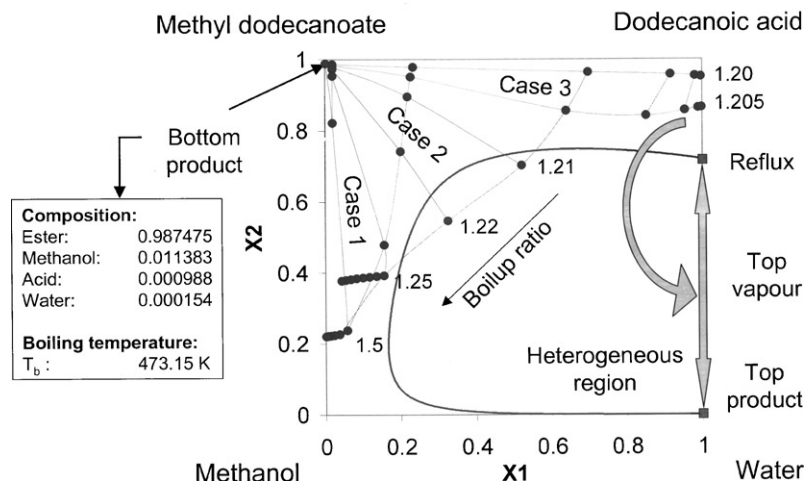


Fig. 13. Concentration profiles in an RD column for the esterification of dodecanoic acid with methanol. Case 1. Trajectory leading to a binary mixture methanol/ester; Case 2. Profile ending-up in the heterogeneous region; Case 3. Low boilup ratios: feasible design with liquid profile in homogeneous region, top vapour condensation in heterogeneous region and separation of almost pure water as top product.

below the acid feed. Note that the material balance is not closed.

Several composition profiles are displayed in Fig. 13. At boilup ratios higher than 1.25 (Case 1) the composition on the upward stages converges to a binary mixture of methanol and methyl dodecanoate. At only slightly lower boilup ratios, (Case 2) the composition profile penetrates into the heterogeneous region. After condensation and decantation of the vapour the water-phase will contain a significant amount of alcohol that needs a second recovery column. In the Case 3, at lower boilup ratios below 1.2 the liquid composition profile converges to a binary mixture of dodecanoic acid and water. The vapour phase contains mostly water, small amount of acid but no alcohol. By condensation and decantation the water can be drawn off quantitatively, whereas the organic phase containing over 97 wt% acid is refluxed to the reaction zone. Hence, the RD column can produce ester of high purity in bottom and deliver almost pure water in top (Fig. 13). Because of high purity products the ratio of reactants in feed is practically equimolar, with an extremely low excess of acid or alcohol depending on final impurity specifications. The exact value can be determined by optimisation.

7.4. Thermodynamic experiments

CPE experiments were carried out in the cell as presented in Fig. 4. It has been found that the equilibrium conversion measurements validate the values calculated by using the thermodynamic data available in the standard databank of Aspen Plus.

7.5. Reviewed conceptual design

An important result from the tray-by-tray calculation is that the design must ensure complete alcohol consumption

in the reactive zone. Only dodecanoic acid and water are allowed in the top vapour stream, from which the water can be easily removed by condensation and decantation. A higher number of equilibrium stages is necessary to achieve simultaneous high conversion of the fatty acid and methanol, in this case 14. The column pressure may be atmospheric, but the presence of a certain amount of methanol in the bottom product is still necessary to limit the bottom temperature to 473 K.

Profiles of temperature, concentrations and ester formation rate are presented in Fig. 14. Acid is fed at 420 K as liquid, while methanol is fed at 400 K as vapour. Because the reaction is slightly exothermic, a small increase in temperature is recorded. It may be observed that the maximum reaction rate is located somewhere to the middle of the reaction zone, its position depending on the reflux ratio. An optimum reflux, where both acid and alcohol are completely converted, gives the maximum reaction rate in middle of the reaction zone. Higher reflux ratio leads to an incomplete consumption of acid, which passes as impurity in the bottom product. In this case the maximum reaction rate shifts closer to the bottom of the reaction zone on the contrary, lower reflux ratio gives incomplete consumption of the alcohol, and the maximum reaction rate moves to the top part of the reaction zone.

Contrary to normal distillation, when product purity increases with reflux, in the present case both purity and yield reach maximum at an optimum reflux rate. This phenomenon has been observed and explained previously for other RD processes as methyl acetate (Agreda, Partin, & Heise, 1990; Bessling, Schembecker, & Simmrock, 1997), MTBE (Doherty & Malone, 2001) and ethylene glycol (Chen, Huss, Malone, & Doherty, 2000). Fig. 14 depicts a favourable reaction profile for obtaining high purity ester. At the top of reaction zone, the excess of acid allows almost complete conversion. Similarly, at the bottom of reaction

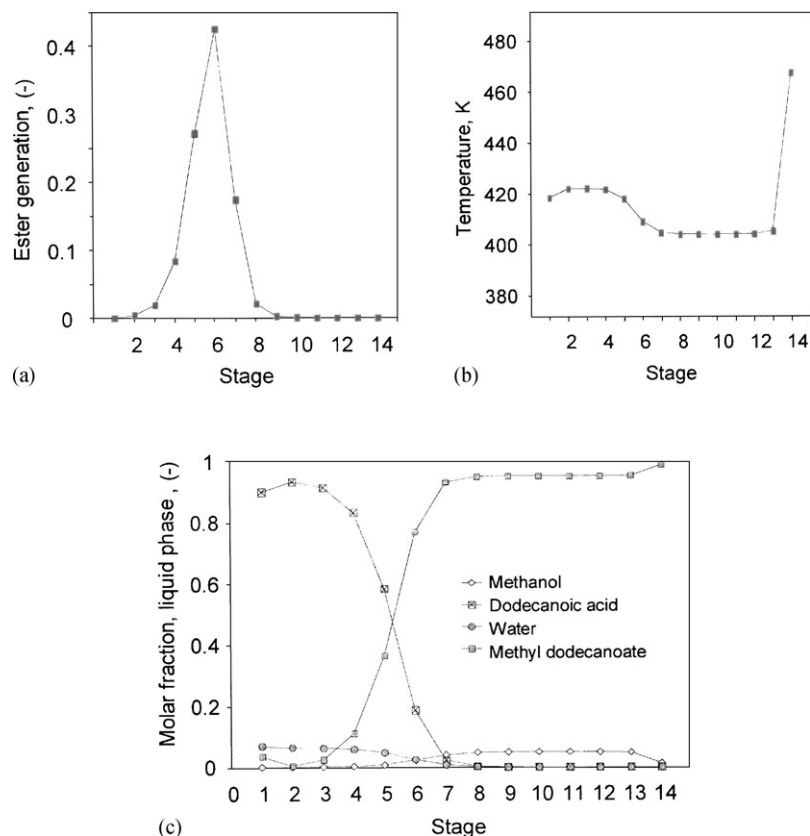


Fig. 14. Profiles in RD column for 2-ethylhexyl dodecanoate manufacture: (a) temperature profile, (b) ester generation profile, and (c) composition profiles.

zone, the local excess of alcohol makes possible the total consumption of the fatty acid.

It was found that reflux ratio plays an important role in controlling the purity (Fig. 13). At higher reflux, the maximum reaction rate moves to the top of the reactive zone, and in opposite direction, by lower values. Both water and ester purities are maximised at optimum reflux ratio, when the highest reaction rate takes place in the middle of reaction zone. It has been found that this optimum reflux rate has a very low value, and operating nearby could give rise to control problems. Operation at a higher value still allows good purity, at least 99 %, still remarkable for such systems. Small reflux values might suggest that the operation is more close to reactive absorption than RD. The temperature in the reactive zone is almost constant, but slightly higher at the feed point of the fatty acid. Thus, VLE seems not to play a role in the temperature profile of the reaction zone. This behaviour is quite different from the esterification of 2-ethylhexanol (see Fig. 11), where the temperature profiles indicated a strong interaction between reaction and separation.

Finally, we may observe that the flowsheet presented in Fig. 2 remains valid in both cases examined, the esterification with 2-ethylhexanol and methanol. As discussed before, by varying the operating parameters, namely the feed temperature, pressure and reflux ratio, the column can be

driven to operate with reflux of alcohol or reflux of acid. The distillate composition is the consequence of operating parameters, by keeping the feed ratio to 1:1. Hence, the alternative corresponding to total consumption of alcohol can be generalised to the esterification reactions of fatty acids with different alcohols by RD.

8. Conclusions

The present research demonstrates that the synthesis of high purity fatty esters is feasible by reactive distillation (RD) using solid catalyst. An RD-set up is proposed for the esterification of the dodecanoic acid with 2-ethylhexanol and methanol, the heaviest and the lightest alcohols in the series C_1 – C_8 . The design ensures selective water removal that shifts the chemical equilibrium to completion, and preserves the catalyst activity.

In the first place the feasibility problem must be solved. The research proposes a methodology that avoids inefficient simulation or laboratory experiments, and may be applied to similar systems. The approach consists of an evolutionary search of the feasible design space by a systematic combination of thermodynamic analysis and computer simulation. Computer investigation sets targets for minimum experimental research.

In a preliminary stage a conceptual design can be developed based on the assumption of chemical equilibrium. Sensitivity analysis demonstrates that the reliability of phase and chemical equilibrium data is essential. In this case additional phase equilibria experiments were necessary.

RCM and ∞/∞ analysis indicate that the esterification of dodecanoic acid with 2-ethylhexanol is feasible at temperatures higher than 373 K necessary for preserving homogeneous conditions. A suitable reaction temperature is around 430 K. Only few equilibrium stages are necessary to achieve high conversion and purity. Water distillates in top with alcohol as heterogeneous azeotrope, and is removed after condensation from the system in the reflux drum by L–L separation. The ester obtained in bottom is diluted with some alcohol to prevent the occurrence of excessive temperatures.

By the esterification of dodecanoic acid with methanol, the ∞/∞ analysis in RCM suggests that the process would not be feasible with alcohol reflux. On the contrary, plate-by-plate calculations demonstrate that the process is feasible with acid reflux. Low reflux ratio is necessary to obtain purity over 99.9%. The second alternative with acid reflux appears to be generic for any type of alcohol.

Notation

A, B, C, D, E, F	coefficients for Antoine equation
A_{ij}, B_{ij}, C_{ij}	binary parameters for UNIQUAC
H	liquid holdup, kmol
k	integration step
K	chemical equilibrium constant
K_γ	correction factor based on the liquid activity coefficients
P	pressure, Pa
q, r	surface and volume parameters for UNIQUAC
S	solubility, wt%
t	time, s
T	temperature, K
V	vapour distillate, kmol s ⁻¹
x	liquid molar fraction
X_1, X_2	transformed coordinates defined as sum of two molar fractions: one reactant and one product. Thus, they are invariable with the extent of reaction
y	vapour molar fraction
$\Delta^f G$	Gibbs free energy of formation, J mol ⁻¹
$\Delta^r G$	Gibbs free energy of reaction, J mol ⁻¹

Greek letters

γ	liquid-phase activity coefficient
θ	dimensionless warped time
ν	stoichiometric coefficient of reaction
ξ	extent of reaction

τ_{ij}	binary interaction parameter for UNIQUAC
ω	molar fraction of liquid phase L_1 in a heterogeneous mixture

Subscripts

a	liquid activity
aq	aqueous phase
az	azeotrope
c	transformed variables/coordinates: 1 for acid + water, and 2 for acid + ester
f	fugacity, Pa
i, j	components in the mixture
org	organic phase
m	liquid phases
T	temperature, K
x	molar fractions in liquid phase

Superscripts

0	at normal pressure, $P = 101.32$ kPa
eq	at CPE
k	iteration step
L	liquid phase
sat	saturation, VLE of pure component

Acknowledgements

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Appendix A. Methodology

A.1. Problem definition and data generation

The following aspects are analysed:

- Feedstock and product specifications with respect to purity.
- Formulation of constraints, as safety and chemical pollution.
- Catalyst selection: properties and potential activity.
- Physical properties of pure components and mixtures, collection of experimental data, as well as estimation by computer methods.

A.2. Preliminary chemical and phase equilibria

This step includes the following aspects:

- Modelling of phase equilibrium (homogeneous and/or heterogeneous).
- Modelling of chemical equilibrium and computation of achievable conversion.

- Computation of distillation boundaries and plot of RCM.
- Identification of reactive azeotropes.

A.3. Equilibrium design

A first evaluation of the flowsheet synthesis is performed on the basis of phase and reaction equilibrium. This step can be decomposed in the following actions:

- Specify the product composition and limiting temperatures. Select the pressure.
- Represent the process in RCM and estimate the feasibility (∞/∞ analysis).
- If the step above fails, check finite reflux feasibility by means of tray-by-tray calculations.
- Determine column configuration and execute rigorous simulation.
- Analyse data uncertainty by means of sensitivity analysis.

The preliminary design is revisited taking into account the improved knowledge on thermodynamics and reactions. Navigating between thermodynamic analysis, design and thermodynamic experiments generates a number of design alternatives.

A.4. Thermodynamic experiments

The preliminary conceptual design allows the identification of additional thermodynamic experiments. Typically these regard:

- Physical properties of pure components.
- Phase equilibria for binaries of significance in design.
- Simultaneous CPE.

Note that experiments are restricted into the space of the feasible design.

A.5. Kinetic experiments

The alternatives from the equilibrium design are ranked. The experimental range of temperature and concentrations is determined. Kinetic experiments should study:

- Kinetics of the un-catalysed and catalysed reaction.
- Selectivity as function of process variables.
- Deactivation rate and mechanism.
- Define tasks for catalyst improvement, or/and search new catalyst.

A.6. Kinetic design

Reaction rate data are inserted in the rigorous simulation model. Sizing elements are determined as the number of stages or of equivalent packing, catalyst loading, hydraulic design and pressure drop, profiles of concentration and tem-

perature, and energy requirements. The final design, as well the final catalyst is selected from alternatives.

A.7. Optimisation

Design optimisation addresses namely pressure and temperature profiles, number of stages and catalyst loading, as well as the reflux ratio and possibly the internal flow rates of some species. Here control elements can be also included.

Appendix B. Tracing RCM with simultaneous CPE

The graphical representation used in this work has the special feature to describe the simultaneous CPE in the both heterogeneous and homogeneous regions. The four components mixture can be represented in only two co-ordinates by means of transformed variables $X_1 = (x_{\text{acid}} + x_{\text{water}})$ and $X_2 = (x_{\text{acid}} + x_{\text{ester}})$. A point in the X_1 – X_2 space is invariant with respect to the extent of reaction.

Finding two-liquid phases at equilibrium could be solved in principle by Gibbs free energy minimisation by searching the condition for two-phase occurrence, but the computation is very demanding (Wasykiewicz & Ung, 2000). We adopted here a simpler but reliable strategy. We start with the VLLE of the heterogeneous *non-reactive* binary mixture acid–water (Eqs. (B.1)–(B.3)). Using a continuation algorithm, we extend the investigation of the heterogeneous region in the *reactive* domain (Eqs. (B.4)–(B.8)). Note that the initialisation of computation is independent of the alcohol type. The computation stops when the LLE converges to the plait point, if the alcohol is completely miscible with water, or to the non-reactive binary alcohol–water in other cases. The procedure continues by tracing RCM in the homogeneous or heterogeneous domains starting from the equilibrium composition determined before. The steps are explained below.

B.1. VLLE for the binary acid–water

There are five unknowns, two components ($i = 1, 2$) in two liquid phases ($m = 1, 2$) plus the temperature, which can be determined from the following equations:

$$\sum_{i=1}^2 x_i^{L_m} = 1 \quad m = 1, 2, \quad (\text{B.1})$$

$$x_i^{L_1} \gamma_i^{L_1} = x_i^{L_2} \gamma_i^{L_2} \quad i = 1, 2, \quad (\text{B.2})$$

$$\sum_{i=1}^2 (x_i^{L_1} \gamma_i^{L_1} P_i^{\text{sat}}) = P. \quad (\text{B.3})$$

B.2. VLLE for reactive mixture of four components

The calculation starts with the temperature and previous compositions as initial guesses. A step of 0.001 in

transformed coordinates is selected. There are nine unknowns: eight molar fractions for each liquid phase, and temperature. The non-linear system of equations is

$$\sum_{i=1}^4 x_i^{L_m} = 1 \quad m = 1, 2, \quad (\text{B.4})$$

$$x_i^{L_1} \gamma_i^{L_1} = x_i^{L_2} \gamma_i^{L_2} \quad i = 1, \dots, 4, \quad (\text{B.5})$$

$$\sum_{i=1}^4 (x_i^{L_1} \gamma_i^{L_1} P_i^{\text{sat}}) = P, \quad (\text{B.6})$$

$$K_a = \prod_{i=1}^4 (x_i^{L_1} \gamma_i^{L_1})^{v_i}, \quad (\text{B.7})$$

$$\sqrt{(X_1^{k+1} - X_1^k)^2 + (X_2^{k+1} - X_2^k)^2} = 0.001, \quad (\text{B.8})$$

where X_c^k represent the transformed composition in the coordinate $c = 1, 2$ at the iteration k .

B.3. RCM in homogeneous liquid phase

The calculation starts at the boundary of the heterogeneous region. The residue curve of a simple differential distillation can be described as function of *warped time* variable $\theta = Vt/H$ with the relation:

$$dx/d\theta = x_i^{\text{eq}} - y_i \quad i = 1, \dots, 4. \quad (\text{B.9})$$

V being the vapour distillate, H the liquid holdup in the distillation still, and t the time. Vapour composition is related to the liquid equilibrium composition by:

$$y_i = \frac{x_i^{\text{eq}} \gamma_i P_i^{\text{sat}}}{P} \quad i = 1, \dots, 4. \quad (\text{B.10})$$

At each integration step of the Eq. (B.9), the liquid composition becomes:

$$x_i^{(k+1)} = x_i^{\text{eq}(k)} + \dot{x}_i^{(k)}(d\theta) \quad i = 1, \dots, 4. \quad (\text{B.11})$$

Because of distillation, the residual liquid in pot is not at chemical equilibrium. However, the position of the point remains the same in transformed co-ordinates X_1 – X_2 . The resulting composition is brought again at CPE,

$$x_i^{\text{eq}} = x_i + \xi v_i \quad i = 1, \dots, 4. \quad (\text{B.12})$$

The extent of reaction and the equilibrium temperature are obtained by solving the system:

$$K_a = \prod_{i=1}^4 (x_i^{\text{eq}} \gamma_i)^{v_i}, \quad (\text{B.13})$$

$$\sum_{i=1}^4 (x_i^{\text{eq}} \gamma_i P_i^{\text{sat}}) = P. \quad (\text{B.14})$$

B.4. RCM in heterogeneous region

The computation integrates Eq. (B.9) starting from a point on the L–L boundary, but in opposite direction. The liquid

of composition x_i^{eq} is split into two phases at CPE with the compositions $x_i^{L_1}$ and $x_i^{L_2}$ ($i = 1, \dots, 4$) by means of Eqs. (B.4)–(B.7). In transformed variables the L–L split is described by:

$$X_c^{\text{eq}} = \omega X_c^{L_1} + (1 - \omega) X_c^{L_2} \quad c = 1, 2, \quad (\text{B.15})$$

where ω is the molar fraction of L_1 . Eliminating ω between the two equations (B.15) leads to a single equation that replaces Eq. (B.8). Once temperature and composition of one liquid phase known, the vapor composition is calculated according Eq. (B.10). Further, the liquid composition at step $(k+1)$ is obtained by Eq. (B.11). The procedure repeats until the liquid composition reaches an unstable/stable node or crosses the LLE boundary back in the homogeneous region.

References

- Agreda, V. H., Partin, L. R., & Heise, W. H. (1990). High purity methyl acetate via reactive distillation. *Chemical Engineering Progress*, 86(2), 40–46.
- Aspen Plus (2001). (release 10.2.1). User Manual.
- Barbosa, D., & Doherty, M. F. (1987). A new set of composition variables for the representation of reactive phase diagrams. *Proceedings of the Royal Society of London A*, 413, 459–464.
- Bessling, B., Schembecker, G., & Simmrock, K. H. (1997). Design of processes with reactive distillation line diagrams. *Industrial & Engineering Chemistry Research*, 36, 3032–3042.
- Bock, H., Wozny, G., & Gutsche, B. (1997). Design and control of a reaction distillation column including the recovery system. *Chemical Engineering Progress*, 36, 101–109.
- Castier, M., Rasmussen, P., & Fredenslund, A. (1989). Calculation of simultaneous chemical and phase equilibria in nonideal systems. *Chemical Engineering Science*, 44(2), 237–248.
- Chen, F., Huss, R. S., Malone, M. F., & Doherty, M. F. (2000). Simulation of kinetic effects in reactive distillation. *Computers and Chemical Engineering*, 24, 2457–2472.
- Cisneros, E. S. P., Gani, R., & Michelsen, M. L. (1997). Reactive separation systems—I. Computation of physical and chemical equilibrium. *Chemical Engineering Science*, 52(4), 527–543.
- Cisneros, E. S. P., Schenk, M., & Gani, R. (1996). Aspects of simulation, design and analysis of reactive distillation column operations. *Computers and Chemical Engineering*, 20(Suppl.), S267–S272.
- Constantinou, L., & Gani, R. (1994). New group-contribution method for estimating properties of pure compounds. *A.I.Ch.E. Journal*, 40(10), 1697–1710.
- Doherty, M. F., & Malone, M. (2001). *Conceptual design of distillation systems*. New York: McGraw-Hill.
- Gmehling, J., Menke, J., Krafczyk, J., & Fischer, K. (1994). *Azeotropic data. Part I*. Weinheim: VCH.
- Joback, K. G., & Reid, R. C. (1987). Estimation of pure-component properties from group-contributions. *Chemical Engineering Communications*, 57, 233.
- Lee, L. S., & Liang, S. J. (1998). Phase and reaction equilibria of acetic acid-1-pentanol-water-n-amyl acetate system at 760 mm Hg. *Fluid Phase Equilibria*, 149, 57–74.
- Lee, L. S., & Lin, R. G. (1999). Reaction and phase equilibria of esterification of isoamyl alcohol and acetic acid at 760 mm Hg. *Fluid Phase Equilibria*, 165, 261–278.
- Okasinski, M. J., & Doherty, M. F. (2000). Prediction of heterogeneous reactive azeotropes in esterification systems. *Chemical Engineering Science*, 55, 5263–5271.
- Sandler, S. (1999). *Chemical and engineering thermodynamics* (3rd ed.). New York: Wiley.

- Stephenson, R., Stuart, J., & Tabak, M. (1984). Mutual solubility of water and aliphatic alcohols. *Journal of Chemical Engineering Data*, 29, 287–290.
- Tuchlenski, A., Beckmann, A., Reusch, D., Duessel, R., Weidlich, U., & Janowsky, R. (2001). Reactive distillation-industrial applications, process design & scale-up. *Chemical Engineering Science*, 56, 387–394.
- Ung, S., & Doherty, M. F. (1995a). Theory of phase equilibria in multireaction systems. *Chemical Engineering Science*, 50(20), 3201–3216.
- Ung, S., & Doherty, M. F. (1995b). Synthesis of reactive distillation systems with multiple equilibrium chemical reactions. *Industrial and Engineering Chemistry Research*, 34, 2555–2565.
- Wasylikiewicz, S. K., & Ung, S. (2000). Global phase stability analysis for heterogeneous reactive mixtures and calculation of reactive liquid–liquid and vapour–liquid–liquid equilibria. *Fluid Phase Equilibria*, 175, 253–272.
- Xiao, W., Zhu, K., & Yuan, W. (1989). An algorithm for simultaneous chemical and phase equilibrium calculation. *A.I.Ch.E. Journal*, 35(11), 1813–1820.